BLI-223 ENGLISH MEDIUM

ORGANISED AND MANAGING INFORMATION

BLI-223: ORGANISING AND MANAGING INFORMATION
TUTOR MARKED ASSIGNMENT

Coverage: **Course Code: BLI-223**
Course: Organising and Managing Information Assignment Code:AST/TMA/ Jul.2025/Jan.2026
Blocks: 1 to 4
Units: 1 to 14 **Total Marks: 70**

Note: Answer all questions.

I) Answer all the questions in not more than 500 words each. (4X10= 40 Marks)

1. Explain the rules in AACR-2R cataloguing of non-book materials. (10)
2. Describe the special features of the electronics version of the DDC. (10)
3. Discuss the canons for arrays. (10)
4. What is a MARC Record? Discuss the format of 10 MARC 21 structure. (10)

II) Answer the following questions in 250 words each. (6X5= 30 Marks)

5) Explain the features and advantages of OPACs. (5)
6) Describe the qualities of Notation. (5)
7) Define the inner forms of a library catalogue. (5)
8) Discuss the Principle of context Dependency. (5)
9) What is a KOS? Discuss its characteristics. (5)
10) Describe the Semantic Web Activities of W3C. (5)

Question 1

Explain the rules in AACR-2R cataloguing of non-book materials.

Answer:

The **Anglo-American Cataloguing Rules, Second Edition, Revised (AACR-2R)** is a widely used standard for the bibliographic description of library materials. While AACR-2R was initially developed for books, it includes specific rules for cataloguing **non-book materials** such as maps, audio-visual media, computer files, manuscripts, and microforms.

Non-book materials often have unique physical characteristics and formats, requiring special treatment for accurate representation in catalogs. AACR-2R provides structured guidance to ensure consistency, clarity, and user accessibility.

Key Elements in AACR-2R for Non-Book Materials

1. **Title and Statement of Responsibility Area (1.1):**
   The title proper is transcribed exactly as it appears on the item. For non-book materials, the title is often taken from labels, containers, or accompanying documentation. Statement of responsibility includes creators, producers, or artists, e.g., directors, composers, illustrators.
2. **Edition Area (1.2):**
   Indicates version or edition, if any. This is especially important for software or instructional media which may have multiple editions.
3. **Material (or Type of Publication) Specific Details Area (1.3):**
   This is unique to non-book items and gives physical details like playing speed (e.g., 33⅓ rpm), colour (e.g., colour/sound), or scale (for maps).
4. **Publication, Distribution, etc. (Imprint) Area (1.4):**
   Includes the place of production, name of the producer, and date. For films or digital media, the production company is treated as the publisher.
5. **Physical Description Area (1.5):**
   This includes the number of physical units (e.g., 1 CD-ROM), specific dimensions (e.g., 4¾ in.), and accompanying material (e.g., booklet, case).
6. **Series Area (1.6):**
   If the non-book material is part of a series, the title and number of the series should be noted.

7. **Notes Area (1.7):**
   Notes provide supplementary information such as system requirements (for digital media), language of content, access restrictions, or technical specifications.
8. **Standard Number and Terms of Availability (1.8):**
   Includes identifiers like ISBN, ISSN, or publisher codes.

Example (for a DVD):

Title: *Gandhi* / directed by Richard Attenborough
Edition: Special collector's edition
Imprint: Mumbai: Shemaroo Entertainment, 2005
Physical Description: 1 videodisc (191 min.): sound, color; 4¾ in. + 1 booklet
Notes: Originally released as a motion picture in 1982. English with Hindi subtitles
ISBN: 1234567890

Challenges in Cataloguing Non-Book Materials:

- **Inconsistent Titles:** Titles may appear on the label, container, or accompanying material, requiring judgement to select the most appropriate one.
- **Physical Complexity:** Items may come with manuals, cases, or multiple pieces needing accurate documentation.
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- **Rapid Format Changes:** Digital formats and multimedia evolve quickly, requiring libraries to stay updated on standards.

Conclusion:

AACR-2R provides comprehensive cataloguing guidelines for non-book materials, ensuring that they are integrated into library catalogs effectively. Accurate description and access points allow users to locate and use these resources just as easily as books.

Question 2

Describe the special features of the electronic version of the DDC

Answer:
The **Dewey Decimal Classification (DDC)** system is one of the most widely used library classification systems in the world. With the growth of digital

technology, an **electronic version of DDC**, known as **WebDewey**, has been developed by OCLC (Online Computer Library Center) to enhance accessibility, efficiency, and usability in the digital age.

Special Features of the Electronic Version of DDC (WebDewey):

1. **Web-Based Access:**
   WebDewey is accessible online via subscription. This eliminates the need for physical volumes, allowing librarians to access it from any location with internet access.
2. **Searchable Interface:**
   The electronic version offers powerful **search tools**, including keyword, number, phrase, and Boolean searches, enabling faster and more flexible retrieval of classification numbers.
3. **Linked Terms and Notes:**
   Terms are hyperlinked, allowing users to quickly jump between related concepts. Notes, including "see also" references and scope notes, are also clickable.
4. **Integrated Authority Control:**
   WebDewey integrates **Library of Congress Subject Headings (LCSH)**, allowing cataloguers to align classification numbers with subject authorities.
5. **User Customization:** .com
   Users can create **custom lists**, add annotations, or save frequently used classification numbers for future use.
6. **Regular Updates:**
   The electronic DDC is updated more frequently than the print version. This ensures classification remains current with new topics and emerging fields.
7. **Multilingual Support:**
   Some versions include translations in languages like French, Spanish, or German, broadening accessibility in multilingual environments.
8. **Hierarchical Browsing:**
   Users can browse the hierarchical structure of DDC—from main classes to subdivisions—facilitating understanding and navigation.

9. **Export and Integration Options:**
   Classification data can be exported or integrated into cataloguing software, streamlining technical services in libraries.
10. **Training Tools and Help Menus:**
    Online help, tutorials, and support features assist users in understanding the interface and best practices for classification.

Advantages Over Print Version:

- **Speed:** Quick retrieval of class numbers through search functions.
- **Accuracy:** Minimizes human error in locating or interpreting classifications.
- **Efficiency:** Saves time in cataloguing and training new staff.
- **Currency:** Always reflects the latest edition and updates without reprinting.

Use in Libraries:

- Academic and research libraries use WebDewey for classifying complex or interdisciplinary subjects.
- Public libraries use it for quick cataloguing of new materials.
- Library schools use WebDewey as a teaching tool for classification.

Conclusion: www. .com

The electronic version of DDC is a powerful tool that modernizes classification processes. By offering enhanced searchability, customization, and integration, it significantly improves the workflow for cataloguers and ensures that libraries can adapt to evolving information landscapes.

Question 3:

Discuss the canons for arrays.

Answer:
In the context of library classification theory, especially as developed by S. R. Ranganathan in his Colon Classification system, "canons" are fundamental principles or rules that govern the formation of classifications. One of the key areas where canons apply is in the construction of arrays.

An array is a group of coordinate classes, i.e., classes that stand on the same level in a hierarchy under a common broader class. For example, the subjects "Physics," "Chemistry," and "Biology" are arrays under the broader subject of "Science."

The canons for arrays aim to ensure that these groupings are logical, exhaustive, and mutually exclusive. The major canons include:

1. Canon of Exhaustiveness:
This canon states that all possible coordinate classes under a given class should be included. An incomplete array can lead to misclassification or confusion. For example, listing only "Physics" and "Chemistry" under "Science" would be incomplete without "Biology" or other related sciences.

2. Canon of Exclusiveness:
Each class in an array must be mutually exclusive, meaning there should be no overlap among them. If a subject fits under more than one class, it leads to redundancy or misplacement. Each concept should belong to one class only within that array.

3. Canon of Co-ordination:
All classes in an array should be of the same order or level in the classification hierarchy. They should be coordinate in status, not subordinate to each other. For example, "India," "China," and "Japan" can form an array under "Asian Countries," but "Delhi" should not be included as it is a city, not a country.

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4. Canon of Succession:
This canon governs the logical and helpful sequence in which the classes in an array are arranged. They can be sequenced alphabetically, chronologically, geographically, or based on increasing complexity depending on the subject matter.

5. Canon of Modulation:
Involves the principle that each class should be clearly distinguishable from its neighboring class by a definite difference in its characteristics. It ensures that classes are not only distinct but also sufficiently defined to prevent confusion.

6. Canon of Helpful Sequence:
It suggests that the arrangement should be in such a way that it helps the user to easily locate the required class. For instance, in a library catalog, placing frequently used classes earlier might help in retrieval.

7. Canon of Comprehensiveness:
This implies that each class should cover all aspects that logically fall within its scope without spilling over into the domain of another class.

8. Canon of Homogeneity:
All classes in an array should be described using the same kind of characteristics or facets. Mixing different types of criteria (e.g., geography with subject) in one array can confuse users.

In summary, the canons for arrays are essential for creating a structured, logical, and user-friendly classification system. They promote clarity, consistency, and precision in the organization of knowledge, which ultimately enhances retrieval and user satisfaction in libraries.

Question 4:

What is a MARC Record? Discuss the format of 10 MARC 21 structure.

Answer:
MARC stands for **MAchine-Readable Cataloging**. It is a standardized format developed by the Library of Congress for the representation and exchange of bibliographic data in machine-readable form. MARC records are vital for enabling library systems to share, interpret, and manage bibliographic data consistently.

What is a MARC Record?
A MARC record is a digital representation of a library item containing metadata fields such as author, title, publisher, subject headings, and classification numbers. MARC makes this information interpretable by computers and allows interoperability between library systems.

Each MARC record consists of three components:

1. **Leader** – 24-character fixed field that provides information for processing the record.

2. **Directory** – Lists the tags and their positions.

3. **Variable Fields** – Composed of data fields divided into subfields.

MARC 21 Format Structure (10 key tags):

1. **Tag 001 – Control Number**
   A unique number assigned by the cataloguing agency (e.g., OCLC number).

2. **Tag 008 – Fixed-Length Data Elements**
   Includes coded data such as date of publication, language, place of publication, etc.

3. **Tag 020 – ISBN**
   Provides the International Standard Book Number for the publication.

4. **Tag 100 – Main Entry – Personal Name**
   Used for author names. Includes first and last name, with subfield indicators for dates and affiliations.

5. **Tag 245 – Title Statement**
   Includes the title proper, remainder of title, statement of responsibility (author, editor), etc.

6. **Tag 260 or 264 – Publication Details**
   Shows the place of publication, publisher name, and date of publication.

7. **Tag 300 – Physical Description**
   Includes information about pagination, illustrations, dimensions, and accompanying materials.

8. **Tag 490 – Series Statement**
   Used to record series information, such as volume or number in a series.

9. **Tag 650 – Subject Added Entry – Topical Term**
   Indicates subject terms using controlled vocabularies like LCSH.

10. **Tag 700 – Added Entry – Personal Name**
    Provides access to additional authors, editors, or contributors not listed in Tag 100.

MARC 21 also includes indicators and subfields within each tag:

- **Indicators:** Two-character codes that provide further interpretation of the tag.

- **Subfields:** Prefixed by a delimiter (often ‡ or $), each holds a specific element of the data.

In conclusion, MARC 21 provides the framework for consistent bibliographic description and data sharing across libraries. Understanding its structure is crucial for information professionals in the digital age.

Question 5:

Explain the features and advantages of OPACs.

Answer:
An OPAC (Online Public Access Catalogue) is an electronic system that provides users with searchable access to a library's catalog through an online interface. It serves as a digital replacement for traditional card catalogs and is integral to modern Integrated Library Management Systems (ILMS).

Features of OPAC:

1. **Search Options:** Users can search by title, author, subject, keyword, call number, and other criteria.

2. **Advanced Search:** Allows Boolean operators, filters (date, language, format), and limiters.

3. **Availability Information:** Shows real-time status of documents – whether available, checked out, or reserved.

4. **User Account Access:** Allows users to log in, renew items, place holds, and view their borrowing history.

5. **Multilingual Interface:** Some OPACs support multiple languages.

6. **Remote Access:** Available via web browsers from any location with internet.

7. **Hyperlinked Records:** Enables users to explore related materials or subject headings.

8. **Integration with Multimedia:** Can provide images, e-books, audio, and video files.

Advantages of OPAC:

- **User-Friendly:** Intuitive interface makes it accessible even to non-technical users.

- **Efficient Retrieval:** Enables quick and efficient search and access to materials.
- **Real-Time Updates:** Information is current, with updates on new arrivals or book status.
- **Accessibility:** Can be accessed 24/7 from anywhere.
- **Cost-Effective:** Reduces printing costs and physical space used by card catalogs.

In essence, OPACs modernize library access and empower users to independently explore resources with ease and efficiency.

Question 6:

Describe the qualities of Notation.

Answer:
Notation in library classification refers to a system of symbols—typically numbers, letters, or a combination—that represents classes and sub-classes in a classification scheme. Notation serves as a shorthand for the subject content of documents and facilitates the physical arrangement and retrieval of materials in libraries.

Key Qualities of Good Notation:

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1. **Simplicity and Brevity:**
   A good notation should be easy to understand and write. It should be as brief as possible while conveying the full meaning. Overly complex notations can confuse users and complicate shelving and retrieval.

2. **Expressiveness:**
   Expressiveness means the notation should reflect the hierarchical structure and relationships between subjects. For example, in the Dewey Decimal Classification, 500 represents Science, and 510 represents Mathematics, which is a subdivision. This helps users understand subject relationships intuitively.

3. **Flexibility:**
   Notation must allow for the insertion of new topics without disrupting the existing structure. Decimal notations (like in DDC) are flexible as they allow for endless subdivisions using decimal points.

4. **Hospitality:**
   This refers to the ability of the notation system to accommodate future developments in knowledge. A notation should be expandable to incorporate new subjects or sub-fields as they emerge.

5. **Mnemonic Value:**
   Where possible, notation should suggest the subject it represents. For instance, using "BIO" for Biology can help recall. Though this is more common in alphabetical systems, some numerical notations use mnemonic devices through standard associations.

6. **Mechanical Arrangement:**
   A good notation allows for easy mechanical or computerized sorting. For instance, numeric notations like DDC can be sorted efficiently by computer systems for shelf arrangement or database queries.

7. **Uniqueness:**
   Each subject or class should have a unique notation to avoid confusion. Ambiguity in notational assignments can lead to misclassification and retrieval errors.

In summary, the effectiveness of a classification system heavily depends on the strength of its notational system. A well-constructed notation helps maintain order, supports retrieval, and enhances the usability of library collections.

Question 7:

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Define the inner forms of a library catalogue.

Answer:
The term "inner forms" of a library catalogue refers to the structure or arrangement of the catalogue entries—the way information is presented and accessed internally by users. Unlike "outer forms," which describe the physical manifestation (cards, books, or digital), inner forms deal with the logical organization of bibliographic records.

Main Inner Forms of a Library Catalogue:

1. **Alphabetical Catalogue:**
   In this form, entries are arranged alphabetically by author, title, or subject. It can be divided into:

- **Author Catalogue**
- **Title Catalogue**
- **Subject Catalogue**
  Alphabetical catalogues are user-friendly and widely used due to their simplicity.

2. **Classified Catalogue:**
   Here, entries are arranged according to a classification system such as DDC or UDC. Each document is assigned a class number, and records are listed in class number order. This form helps users browse resources on related topics together and is useful for subject-specific searches.

3. **Alphabetico-Classed Catalogue:**
   This is a hybrid form that combines both classified and alphabetical features. The main entries are arranged alphabetically under broad subject headings, which are themselves arranged in a classified manner. It attempts to harness the advantages of both systems.

4. **Dictionary Catalogue:**
   In this form, all entries—author, title, subject, and added entries—are listed in a single alphabetical sequence, much like a dictionary. This form is convenient and comprehensive for users.

Each inner form has its own advantages depending on the user's need. A well-designed inner form improves the efficiency and accessibility of the library's resources.

Question 8:

Discuss the Principle of Context Dependency.

Answer:

The Principle of Context Dependency is an important concept in library classification theory and information organization. It refers to the idea that the classification or interpretation of a subject or concept is dependent on its context—i.e., the environment or setting in which it occurs or is being studied.

Explanation:
In library classification, the same term may mean different things in different contexts. For instance, the term "mouse" could refer to:

- An animal (in zoology),

- A computer device (in technology), or
- A cartoon character (in entertainment).

Depending on the subject domain, the appropriate classification will vary. Therefore, assigning a class number without considering the context would result in misclassification.

Applications in Classification Systems:

- **Facet Analysis (Ranganathan):** In Colon Classification, context is determined by breaking down a subject into facets such as personality, matter, energy, space, and time.
- **Notation Design:** Some classification schemes use hierarchical notation to reflect context, ensuring that similar concepts are grouped logically.
- **Indexing and Retrieval:** When cataloguing or indexing, understanding the context ensures the subject headings or descriptors are accurate and specific to the user's intent.

Significance:

- **Reduces Ambiguity:** Avoids misinterpretation and helps users find exactly what they are looking for.
- **Improves Relevance:** Enhances the precision of search results and user satisfaction.
- **Supports Interdisciplinary Work:** Recognizes that concepts often span multiple fields and need context-based classification.

In summary, the Principle of Context Dependency highlights the importance of considering the setting or environment in which a subject exists. It ensures accuracy and relevance in cataloguing and classification.

Question 9:

What is a KOS? Discuss its characteristics.

Answer:
KOS stands for **Knowledge Organization System**. It refers to structured tools or frameworks used to organize information and knowledge systematically to facilitate retrieval, access, and use.

Types of KOS include:

- Classification schemes (e.g., Dewey Decimal, UDC)
- Thesauri (e.g., ERIC Thesaurus)
- Subject headings (e.g., LCSH)
- Ontologies
- Taxonomies
- Authority files

Characteristics of a Good KOS:

1. **Structured Vocabulary:**
   A KOS provides a controlled vocabulary that standardizes the terms used to describe concepts, thus reducing ambiguity.

2. **Hierarchical Relationships:**
   Many KOSs use broader-narrower term relationships to organize concepts in a tree-like structure, facilitating top-down or bottom-up searches.

3. **Semantic Relationships:**
   Apart from hierarchy, KOS also defines associative (related term) and equivalence (synonym) relationships among concepts.

4. **Consistency:**
   KOS ensures consistency in indexing and classification across different systems, which is crucial for interoperability and metadata harvesting.

5. **Support for Retrieval:**
   They improve precision and recall in search systems by mapping user queries to standardized descriptors.

6. **Machine-Readability:**
   Modern KOSs are often designed to be machine-readable, enabling integration with information retrieval systems, search engines, and digital libraries.

7. **Dynamic Updating:**
   To remain relevant, a KOS should be able to adapt and update with the evolution of knowledge in various domains.

In conclusion, KOS is fundamental to modern information systems. It bridges the gap between user vocabulary and indexed content, enhances discovery, and supports efficient information management.

Question 10:

Describe the Semantic Web Activities of W3C.

Answer:
The **Semantic Web** is an extension of the existing World Wide Web, aiming to make data on the internet machine-readable and understandable. It enables computers to interpret, connect, and manipulate web data meaningfully. The **World Wide Web Consortium (W3C)** leads the development of Semantic Web standards.

Key Semantic Web Activities by W3C:

1. **Resource Description Framework (RDF):**
   RDF is a foundational framework that represents information about resources on the web using subject-predicate-object triples. It allows structured and semi-structured data to be linked across different platforms.

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2. **Web Ontology Language (OWL):**
   OWL enables the definition of complex relationships and vocabularies. It is used to create ontologies—formal representations of a set of concepts and relationships within a domain.

3. **SPARQL Protocol and RDF Query Language:**
   SPARQL allows users to query RDF datasets, making it easier to retrieve and manipulate semantic data from databases.

4. **Linked Data Initiatives:**
   W3C promotes the use of URIs to connect related data across the web. Projects like DBpedia and Open Government Data portals exemplify this.

5. **SKOS (Simple Knowledge Organization System):**
   SKOS provides a model for expressing the structure and content of

concept schemes such as thesauri, classification schemes, and subject heading systems.

6. **PROV Data Model:**
   Supports the tracking of data provenance, allowing users to trace how a dataset was created and modified.

W3C's Semantic Web activities aim to make web data more intelligent and interconnected, opening the door for advanced applications in AI, libraries, research, and data science.

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